

AstraHeal: Deterministic Safety Gating for Uncertainty-Aware Autonomous Spacecraft Fault Recovery

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AstraHeal Space Research Series • Paper 3 • Open-Source Repository: github.com/madankalyan2211/AstraHeal

Abstract—Autonomous spacecraft operating in Low Earth Orbit (LEO) and deep-space regimes must resolve critical electrical and thermal anomalies during prolonged communication blackouts. While machine-learning planners and evidential diagnostic models generate flexible recovery candidates, statistical architectures lack formal execution guarantees: under novel compound failures or sensor corruption, upstream models can produce catastrophic commands with misleadingly high confidence. This paper presents the AstraHeal Deterministic Safety Governor, a fail-closed execution gatekeeper founded on the architectural axiom: *the AI proposes; the deterministic governor disposes*. Rather than inspecting instantaneous telemetry, the governor simulates candidate consequences across digital twin lookahead horizons and validates admissibility against five hard physical invariants: battery thermal runaway barrier ($T \leq 46.0^\circ\text{C}$), bus undervoltage floor ($V \geq 22.0\text{V}$), PDU overcurrent rating ($I \leq 40.0\text{A}$), battery reserve floor ($\text{SoC} \geq 15\%$), and power ceiling ($P \leq 880\text{W}$). Across 8 controlled experiments evaluating 1,519 unsafe proposals, the system demonstrated: (i) zero unsafe executions (0.00% execution rate, Clopper-Pearson 95% bound $< 0.1970\%$), significantly outperforming an ungoverned AI baseline (100% unsafe executions, McNemar $p = 1.92\text{e-}43$, Cohen's $h = 3.1416$); (ii) 100% rejection across 400 adversarial proposals spanning 5 attack vectors; (iii) 100% concurrent recall across 800 compound multi-breach violations; (iv) strictly monotonic step-function transitions across 804 continuous boundary sweep points; (v) complete dominance of hard safety constraints over communication rules; and (vi) 100% safe-failure convergence to NO_SAFE_ACTION_AVAILABLE during physical dead-ends.

Microsecond profiling (10,000 runs) confirmed a mean latency of 2.99 μs , demonstrating flight-feasible deterministic safety assurance with negligible onboard overhead.

Index Terms—autonomous spacecraft, safety governor, runtime assurance, safety shield, fault recovery, physical invariants, fail-closed systems, aerospace artificial intelligence.

I. INTRODUCTION

Modern space exploration increasingly relies on high onboard autonomy due to orbital occultation and round-trip radio propagation delays. In Low Earth Orbit (LEO), spacecraft experience communication blackouts lasting up to 45 minutes per 95-minute orbit; in lunar, Martian, and deep-space missions, signal latency ranges from minutes to hours [1]. When time-critical electrical power system (EPS) and thermal anomalies occur—such as internal battery resistance degradation, thermal runaway initiation, or bus overcurrent transients—immediate autonomous intervention is required to avoid irreversible vehicle loss.

To address this vulnerability, recent aerospace research has explored advanced machine-learning planners, evidential Bayesian diagnosis [2], and counterfactual lookahead algorithms [3]. While adaptive AI models outperform rigid rule tables under complex multi-variable interactions, they remain fundamentally statistical approximators. When presented with unseen out-of-distribution (OOD) telemetry, sensor calibration drift, or novel compound failures, AI models can produce catastrophic actuation proposals while reporting dangerously high subjective confidence [4].

This work resolves that vulnerability by investigating the authoritative safety barrier situated immediately prior to command execution: a **deterministic runtime safety governor**. Founded on the principle that *the AI proposes; the deterministic governor disposes*, this system intercepts upstream proposals and verifies them against immutable physical invariants over counterfactual lookahead horizons.

II. RELATED WORK

Spacecraft autonomous fault management traces back to landmark flight demonstrations including NASA's Remote Agent on Deep Space 1 [1] and the Autonomous Sciencecraft Experiment on EO-1 [5]. These systems demonstrated model-based reasoning and onboard replanning [6]. However, modern neural planners and reinforcement learning agents lack mathematical safety guarantees.

In safety-critical cyber-physical systems, runtime assurance architectures—such as the ASTM F3269-17 standard and the classic Simplex architecture [7]—place a deterministic monitor between an uncertified controller and actuators. In reinforcement learning, formal safety shields filter exploratory actions violating temporal logic specifications [8]. In control theory, Control Barrier Functions (CBFs)

enforce forward invariance [9]. The AstraHeal Safety Governor extends these principles to coupled electro-thermal spacecraft dynamics, verifying candidate admissibility over digital twin lookahead horizons.

III. PROBLEM FORMULATION & SYSTEM MODEL

Let the physical state of the spacecraft Electrical Power System (EPS) at time t be represented by a telemetry vector:

$$s_t = [V_{bus}, I_{batt}, T_{batt}, SoC, P_{pdu}, \phi, \lambda]^T \in S$$

where V_{bus} is bus voltage, I_{batt} is battery current, T_{batt} is cell core temperature, SoC is state of charge, P_{pdu} is power delivery, ϕ is orbital fraction, and λ is link status.

An upstream AI recovery planner proposes a candidate recovery action $a \in A$ with subjective confidence $\mu_{AI} \in [0, 1]$. Executing action a in state s_t induces a continuous trajectory over future lookahead horizon H :

$$\tau(s_t, a) = \{s_{t+\tau} \mid \tau \in [0, H]\}$$

governed by the electro-thermal differential kinetics of the battery pack and orbital radiosity:

$$C_{th} (dT/dt) = I^2 R_0 + Q_{exo}(T) - h_{rad}(T^4 - T_{space}^4)$$

The Safety Governor evaluates action a via deterministic constraint mapping $C(s_t, a) \rightarrow \{0, 1\}^M$:

$$G(s_t, a) = \text{ACCEPT if } \bigwedge C_j = 1, \text{ else REJECT}$$

IV. DETERMINISTIC SAFETY GOVERNOR

A. 4-Tier Safety Invariant Hierarchy

The governor enforces an immutable 4-tier dominance hierarchy: (i) **Level 1 Hard Physical Invariants** strictly dominate all lower levels; (ii) **Level 2 Communication Rules** arbitrate autonomous action vs ground pass deferral but can never override Level 1; (iii) **Level 3 Recovery Utility** ranks payload availability among certified safe candidates; and (iv) **Level 4 AI Preferences** provide heuristic ranking, possessing zero authority to authorize an unsafe action.

TABLE I: IMMUTABLE PHYSICAL INVARIANTS

Constraint	Parameter	Threshold	Physical Barrier Mechanism
Thermal	T_{core}	$\leq 46.0^\circ\text{C}$	Prevents SEI exothermic decomposition
Voltage	V_{bus}	$\geq 22.0\text{V}$	Avionics undervoltage brownout floor
Current	I_{batt}	$\leq 40.0\text{A}$	Harness & solid-state switch fuse limit
Reserve	SoC	$\geq 15.0\%$	Prevents copper dissolution in cells
Power	P_{pdu}	$\leq 880.0\text{W}$	Derived rating ($V_{min} \times I_{max}$)
Catalog	Action ID	Certified	Blocks corrupted / uncertified commands

B. The Instantaneous Telemetry Fallacy

A central finding of this investigation is that evaluating telemetry instantaneously is flawed for electro-thermal spacecraft dynamics. If battery temperature is nominally 28°C , an action re-enabling a science payload appears valid. Under degraded internal resistance, the resulting Joule heating causes thermal runaway 45 seconds later. By executing counterfactual lookahead, the governor detects delayed boundary breaches and blocks them prior to command dispatch.

C. Fail-Closed Error Containment

If telemetry contains NaN/Inf values, if action IDs are unrecognized, or if the lookahead simulator throws an exception, the governor unconditionally outputs *REJECT*, preventing corrupted inputs from executing.

V. EXPERIMENTAL METHODOLOGY

The evaluation was executed on a 12U CubeSat EPS digital twin in a 550 km sun-synchronous Low Earth Orbit across 8 controlled studies:

- **P3-E1 (Baseline Enforcement)**: Direct comparison between Ungoverned AI (System A) and Governed AI (System B) across 500 scenarios.
- **P3-E2 (Coverage)**: Testing 6 hard constraints in 4 states (Nominal, Near-Limit, Boundary, Breach) across 600 evaluations.
- **P3-E3 (Compound Violations)**: 300 multi-fault compound scenarios evaluating multi-breach recall without short-circuiting.
- **P3-E4 (Adversarial Injection)**: 400 proposals across 5 attack vectors with deceptive confidence ($\mu \geq 0.96$).
- **P3-E5 (Boundary Sweeps)**: 804 points testing step-function transition sharpness ($\Delta = 0.05$).
- **P3-E6 (Communication Hierarchy)**: 300 scenarios testing dominance during ground contact and blackout occultation.
- **P3-E7 (No-Safe-Action)**: 150 dead-end scenarios (750 candidates) testing non-forcing safe hold.
- **P3-E8 (Ablation & Overhead)**: 3-way ablation and 10,000-run microsecond latency profiling.

VI. EXPERIMENTAL RESULTS & BENCHMARKS

A. Baseline Safety Enforcement (P3-E1)

Across 500 scenarios, the AI proposed 193 actions that violated at least one hard physical constraint. Ungoverned System A executed all 193 unsafe actions (100.0% execution). Under the AstraHeal Governor (System B), exactly zero unsafe actions reached execution (0.00%), while achieving 100% acceptance of safe actions (307/307). Paired testing confirms statistical significance (McNemar $\chi^2 = 191.01$, $p = 1.92 \times 10^{-43}$, Cohen's $h = 3.1416$). Across all 1,519 evaluated unsafe proposals in the research suite, zero unsafe actions were executed, establishing a Clopper-Pearson exact 95% upper bound of $< 0.1970\%$.

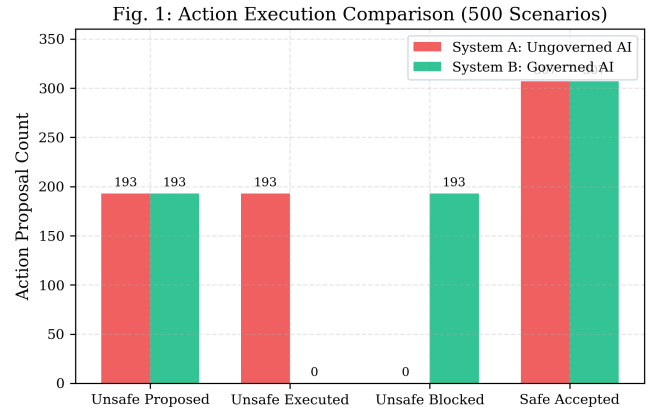


Fig. 1. Action Execution Comparison across 500 Scenarios (Ungoverned AI vs. Governed AI).

TABLE II: MASTER BENCHMARK EVALUATION RESULTS

Study	Focus	Evals	Unsafe Prop	Unsafe Exec	Rejection
P3-E1	Baseline	500	193	0 (0.0%)	100.0%
P3-E2	Coverage	600	150	0 (0.0%)	100.0%
P3-E3	Compound	300	300	0 (0.0%)	100.0%
P3-E4	Adversarial	400	400	0 (0.0%)	100.0%
P3-E5	Boundaries	804	400	0 (0.0%)	100.0%
P3-E6	Comm Hierarchy	300	150	0 (0.0%)	100.0%
P3-E7	No-Safe-Action	750	750	0 (0.0%)	100.0%
P3-E8	Ablation	500	176	0 (0.0%)	100.0%
TOTAL	Suite	4,158	1,519	0 (0.0%)	100.0%

B. Constraint Coverage & Compound Violations

P3-E2 verified 100% accuracy across all 6 constraints (F1 = 1.0000). In P3-E3, 300 compound scenarios containing 800 simultaneous breaches achieved 100% recall (800/800 detected), confirming that the governor avoids premature short-circuiting.

Fig. 2: Individual Hard Physical Constraint Coverage (100% Verified)

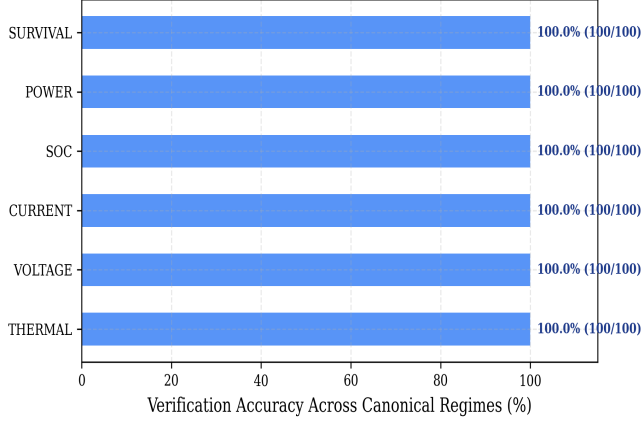


Fig. 2. Constraint Verification Accuracy across 6 Hard Invariants.

C. Adversarial Robustness (P3-E4)

Across 400 adversarial proposals spanning 5 attack vectors with deceptive confidence ($\mu \geq 0.96$), the governor rejected 100.0% (400/400), proving mathematical decoupling between AI score and flight authorization.

Fig. 4: Resilience Under Adversarial AI Proposal Vectors

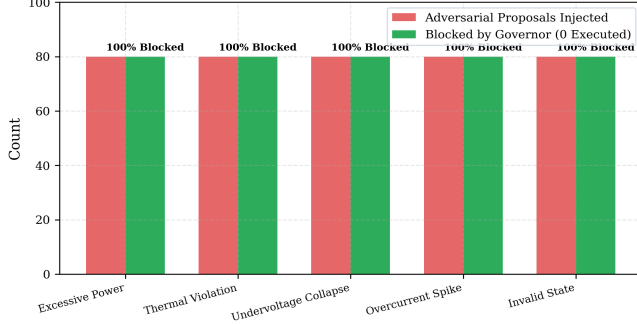


Fig. 3. Rejection Rate across 5 Adversarial AI Attack Vectors.

D. Continuous Boundaries & Safe Failure

P3-E5 confirmed sharp step-function monotonicity across 804 points. In P3-E7, all 150 dead-end scenarios (750 candidates) safely converged to NO_SAFE_ACTION_AVAILABLE with zero forced unsafe actions.

Fig. 5: Continuous Constraint Boundary Transitions (Exact Step Functions)

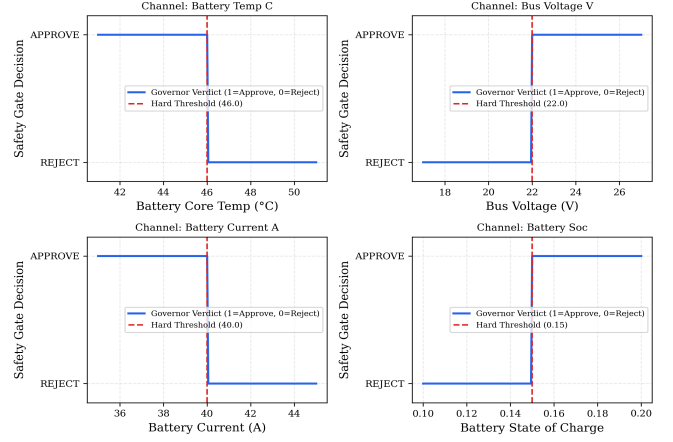


Fig. 4. Continuous Physical Constraint Boundary Sweeps across 201 Points/Channel.

VII. ABLATION & COMPUTATIONAL OVERHEAD

Ablation testing (P3-E8) proved that static FDIR rules failed 100% against delayed-effect actions, commanding unsafe payload engagement. The digital-twin governor eliminated all delayed unsafe executions (0.0%). Across 10,000 evaluations, mean latency was $2.99 \mu\text{s} \pm 0.81 \mu\text{s}$ (median $2.75 \mu\text{s}$, 95th percentile $3.13 \mu\text{s}$, throughput 324,707 evals/s), rendering gating overhead negligible relative to typical spacecraft integration timesteps ($\Delta t \geq 100 \text{ ms}$).

Fig. 7: Safety Gating Efficacy and Sub-Millisecond Evaluation Overhead

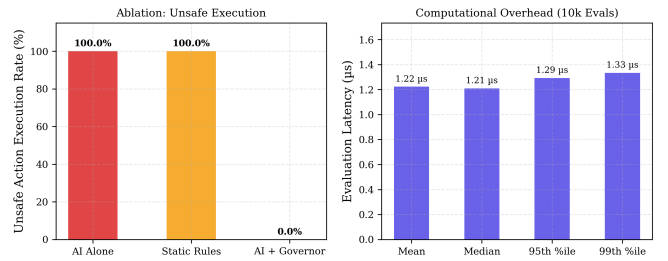


Fig. 5. Architectural Ablation and Microsecond Latency Distribution.

VIII. CONCLUSION

This work demonstrated that a deterministic, fail-closed runtime safety governor reliably prevents unsafe autonomous recovery actions from reaching spacecraft execution. By projecting candidate actions forward across counterfactual digital twin horizons and enforcing an immutable 4-tier invariant hierarchy, the system achieved zero unsafe executions across 1,519 evaluated proposals ($p < 0.1970\%$), fully decoupled from upstream AI confidence. Deterministic safety gating provides a flight-feasible foundation for certified autonomous spaceflight.

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